Cross-Domain Few-Shot Learning Based on Feature Disentanglement for Hyperspectral Image Classification

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***Abstract*— Existing hyperspectral cross-domain few-shot learning (FSL) methods focus mainly on elaborating on training strategies or domain alignment algorithms, while paying less attention to the biased metaknowledge introduced by a large amount of source data and the implicit encouragement of learning target domain-specific attributes. In this article, from the perspective of disentangled representation learning, a novel cross-domain FSL method based on feature disentanglement (FDFSL) is proposed for hyperspectral image classification (HSIC). Specifically, to suppress the representation biased toward the source data and enable the model to implicitly focus on the inherent knowledge of the target domain, an orthogonal low-rank feature disentanglement method is employed to acquire desired features of source and target pipelines. Furthermore, to preserve more shared and discriminative information from the heterogeneous data space (i.e., the spectral dimensions of the source and target scenes are typically different), a multiorder spectral interaction block based on central position encoding (MICD) is proposed to fully integrate the respective features into the spectral domain, which allows the model to emphasize informative spectral dimensions in a data-driven manner. Finally, to diversify the feature representation space while preventing the model overfitting to domain alignment task, a self-distillation scheme is developed to facilitate the acquisition of task-relevant feature components. Extensive experiments and analysis on three public HSI datasets suggest the superiority of the proposed method. The code will be available on the website at https://github.com/Qba-heu/FDFSL.**

***Index Terms*— Cross-domain, feature disentanglement, few-shot learning (FSL), hyperspectral image classification (HSIC).**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain both rich spectral and spatial information, which can reveal physical properties of various land covers [1], [2]. As one of the fundamental guarantees for HSI interpretation, HSI classification (HSIC) plays a key role in real-world equipment of remote sensing [3], [4], [5]. Thanks to the substantially discriminative features incorporated in spectral domains and powerful prior coming from the spatial relationship, HSIC models are able to assign the land cover class for each pixel [6], [7]. Many existing HSIC methods have presented excellent classification performance under supervised learning and introduced a variety of remarkable and fancy architectures specializing in different HSIC needs, such as convolutional neural network (CNN)-based [8], [9], [10], graph neural network (GNN)-based [11], [12], and Transformer-based [13], [14] networks. These methods can perform well on single scenes with sufficient labeled data. However, limited by the low spatial resolution and large coverage area, labeling accurate hyperspectral annotations is extremely challenging and even requires on-site manual expertise [15].

To reduce the cost of repeatedly labeling data, cross-scene HSIC aims to utilize a well-annotated source HSI (source domain) to improve the performance of the classification model on the target HSI (target domain) [16]. Due to the inherent geographical variations in different HSIs, there naturally exist distribution shifts in feature distributions across different hyperspectral scenes, which may lead to models suffering from significant performance degradation when the test distribution differs from the training distribution [17], [18], [19]. To enhance the generalization capability in new HSIC scenes, researchers have devoted their efforts to various learning strategies, including pretraining-based [20] transfer learning and domain adaptation (DA) [21]. These approaches primarily attempt to transfer the abstract or an invariant feature representation of the same category to new datasets. Despite the promising results achieved by these methods, they are weak in handling new scenes with disjoint label spaces

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(i.e., the number and types of categories are different between the source HSI and target HSI) [22] or collected from heterogeneous sensors (i.e., the spectral dimension of the source and target HSIs is different).

Few-shot learning (FSL), the successful application of metalearning in supervised tasks, promises to allow a model to learn novel concepts (classes) from limited experience (samples) [23]. FSL aims at learning a model from previous experiences that can be efficiently adapted to similar yet new tasks only from a few samples [24], [25]. In particular, a model in the metatraining phase first focus on gathering experience across multiple similar metatasks typically from a large training set. In the metatest stage, this model will use metaknowledge to learn new classes from given limited samples. This ignites a few recent striking-back FSL-based research to address the challenging HSIC tasks with limited labeled samples [26], [27], [28]. Most of these works typically select an HSI encompassing a diverse range of categories and a large number of labeled samples or multiple HSIs with distinct land cover classes to construct a training set for reorganizing metatasks. Subsequently, the trained model will be adapted to a new HSI scene for identifying unseen classes from a handful of samples. However, there naturally exists a distributional gap between the metatraining set and the metatest data, meaning that the data in the testing scene is sampled from a distribution that was not encountered during training, resulting in the significant performance drop.

Broadly, some recent approaches [29], [30] try to address this issue by reducing the distribution shift between metatraining and metatest data and pose a new challenge, namely cross-domain hyperspectral FSL (CDFSL). CDFSL not only requires carefully designing training strategies to better learn metaexperiences, but also consideration of the relevant knowledge transfer from the observed distribution to an unseen test distribution during training. Although cross-domain FSL may appear to address cross-scene classification, its setting is distinct from domain adaptation. DA primarily tackles scenarios where the source and target domains share the same classes [31], but the target domain lacks any annotated information. In cross-domain FSL, there is no requirement for a continuous label space between the source and target domains, but the target domain needs to provide N-shot annotated information. Furthermore, cross-scene classification based on transfer learning aims to transfer invariant feature representations to the target scene, while Cross-domain FSL seeks to transfer metaclassification knowledge to the target scene.

A favorite framework is FSL with DA techniques, which constructs metatasks for model training and simultaneously reduce distributional gaps through domain alignment schemes. For example, Li et al. [32] first took the problem of distribution gap in FSL into account and proposed a deep cross-domain FSL for HSIC, which bridges domain gaps by the adversarial DA technique. Zhang et al. [33] proposed a graph information aggregation cross-domain few-shot learning (Gia-CFSL) that incorporates graph information aggregation to achieve domain alignment between the source and target data from both geometric and statistical distribution perspectives during metalearning stage. Although these works can bridge the distribution gap between metatraining and metatest data, such forced and coarse alignment of two domains with disjoint label spaces may far from optimal solution. What is even worse is that, despite aligning the distributions of the two domains, the significant disparity in the amount of data between the source and target domains will unavoidably infuse unfavorable source bias into metalearner, making it challenging for the model to adapt to the target domain. As a result, it is of great importance to explore the additionally shared knowledge that can be leveraged from the target domain, rather than to excessive elaborate more effective training strategies or employ superior domain alignment methods to narrow domain shift.

In addition, due to the potentially inherent sensor discrepancy between source scenes and target scenes, their number of spectral bands may be different. Such a discrepancy brings a heterogeneous gap between source and target feature spaces [34]. The existing CDFSL methods mainly focus on the transfer of homogeneous knowledge and have not specifically addressed the heterogeneity. They either rely on existing band selection algorithms [35] to reduce the data from different dimensions to the same low-dimensional space (in a nontask-driven manner), or they simply use a simple projection layer [32], [33] to map the data to the same feature space (but without any special design). Hence, the exploration of useful knowledge sharing among heterogeneous data remains to be further developed.

To address the aforementioned problems, a novel cross-domain FSL method based on feature disentanglement for HSIC, termed as FDFSL, is proposed in this article. Specifically, to alleviate the bias introduced by the specific characteristic of the source domain, the source feature is orthogonally decoupled as domain-common and domain-specific components, and the common one is adopted to perform the metatask of the source pipeline. To enable the model to implicitly focus on the inherent knowledge of the target domain, the domain-specific features of target domain are employed for constructing metatasks of the target pipeline. Furthermore, to emphasize informative and discriminative spectral bands in a data-driven manner, we propose a multiorder spectral interaction block based on central position encoding (MICD) to fully interact the respective features along spectral domains. To diversify the feature representation space while preventing the model from losing the original semantic attributes caused by the potential overfitting on domain alignment task, a self-distillation scheme is developed to facilitate the acquisition of task-relevant feature components. Additionally, as we have not made specific improvements to the FSL loss and domain alignment loss, FDFSL can be extended to most existing hyerpsectral CDFSL methods. By improving these indispensable components, FDFSL will achieve even better classification performance.

In summary, the main contributions of this article can be summarized as follows.

1) We proposed a cross-domain FSL framework based on feature disentanglement for HSIC, which can suppress the knowledge biased toward the source domain and encourage the model to learn specific information

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from the target domain. An orthogonal low-rank feature disentanglement method is employed to effectively perform feature decoupling, then the source metatasks built by domain-common components will reduce the domain-specific bias and the target metatasks built by domain-specific components will focus on the specific characteristic attribution of the target domain.

2) We design an MICD in the proposed framework, which is capable of exploring more discriminative information from heterogeneous data in a task-driven manner. The fashion of multiorder interactions enables the model to fully uncover underlying patterns in the spectral domain and emphasizes informative bands. The central position encoding (CPE) enables the interaction to sufficiently focus on the correlation of spectral dimensions while taking spatial relationships into consideration.
3) A self-distillation-based feature constraint strategy is developed to incorporate semantic information into embedding features, which can avoid model overfitting to domain alignment tasks and enrich the feature representation for subsequent feature disentanglement tasks. The fusion of features provides a more diverse feature representation space for subsequent feature disentanglement tasks, which can better facilitate the acquisition of expected domain-common or domain-specific feature components.

The remainder of this article is organized as follows. In Section II, the related work is summarized comprehensively. In Section III, the proposed FDFSL is presented in detail. Section IV reports and analyzes experimental results. Finally, conclusions are outlined in Section V.

II. Related Work

A. Few-Shot Learning

FSL research in the field of HSIC can be broadly categorized into two lines: conventional FSL and cross-domain FSL. The former one mainly focuses on exploring algorithms to improve the modeling capability of the relationship between support sets and query sets, enabling the metalearner to better learn desired experiences from source datasets. The latter one establishes a more challenging scenario where the training and testing data are sampled from different distributions, which requires the model to generalize to unseen target datasets with distribution shifts. For conventional FSL methods, they can be roughly categorized into prototype-based [27], relation network-based [36], and siamese network-based [37] approaches. Our work specifically focuses on cross-domain FSL, hence we will mainly provide a brief introduction to the related works in CDFSL.

Due to the inherent geographical variations in different HSIs, there naturally exist distribution shifts in feature distributions across different hyperspectral scenes. Li et al. [32] promote the problem of distribution shift into hyperspectral FSL and proposed a deep CDFSL method. Later, Bai et al. [30] addressed the distribution shift by employing adaptive subspaces transformation that can adaptively transform the feature into a share subspace for FSL. Wang et al. [38] proposed a cross-domain FSL method based on classwise attention, which aims to take the conditional distribution into consideration via category-level attention modules. In addition to cross-domain FSL methods for hyperspectral data, some novel and effective studies in optical images have also been proposed in recent years. For example, inspired by the performance of adapters, Li et al. [39] designed task-specific adapter heads to better adapt the metatrained model to target datasets with distribution shifts and selectively transfer knowledge. Li et al. [40] viewed the FSL task as an image retrieval, which aims to transform features into a nonlinear subspace to minimize task-irrelevant features and adopt a hyperbolic tangent transformation to improve the transferrable discriminative information. Although the paper [40] shares some similarities in motivation with FDFSL, it mainly transforms FSL into image retrieval by designing a new metalearning loss. FDFSL not only uncovers shared metaknowledge by removing source domain bias but also seeks to enhance the learning of target domain-specific knowledge. These methods focus mainly on the designs of few-shot training schemes or domain alignment algorithms, while paying less attention to the source-biased metaknowledge and the implicit target domain-specific attributes.

B. Feature Disentanglement for Representation Learning

Features disentanglement stems from disentangled representation learning, which treats learning as a transform function that is capable of disentangling the explanatory factors of variations behind data and keeping the integrity of relevant information. Additionally, previous feature disentanglement techniques are mainly utilized for domain generalization, where features are decomposed into understandable subfeatures, with one feature being domain-common features and the other domain-specific features. Undoing the dataset bias (UndoBias) [41] and its variants [42], [43] designs different network parameters specialized for expected components to extract their respective domain-invariant and domain-specific features. Taking advantage of generative models, some methods intend to decouple features at the domain level, sample level, and label level from the perspective of sample generation mechanisms [44], [45]. They mainly rely on the design framework of the variational autoencoder (VAE) to learn domain-related information, class-related information, or other information, and they leverage a reconstruction loss to keep basic information complete. In addition, some methods adopt causal learning mechanisms to explore variable relationships (i.e., true cause) from a perspective beyond statistics [46], [47], ensuring that predictions are not influenced by irrelated factors.

These methods typically discard the decomposed domain-specific features and solely focus on learning domain-common features to generalize unseen domains. However, domain-specific features may also contain strong correlates with labels in individual domains and the generalization to other domains. To this end, Bui et al. [48] successfully utilized the relevant information and discarded redundant information within domain-specific features to effectively leverage the helpful domain-specific components for domain generalization tasks. Inspired by these methods, we argue that the source

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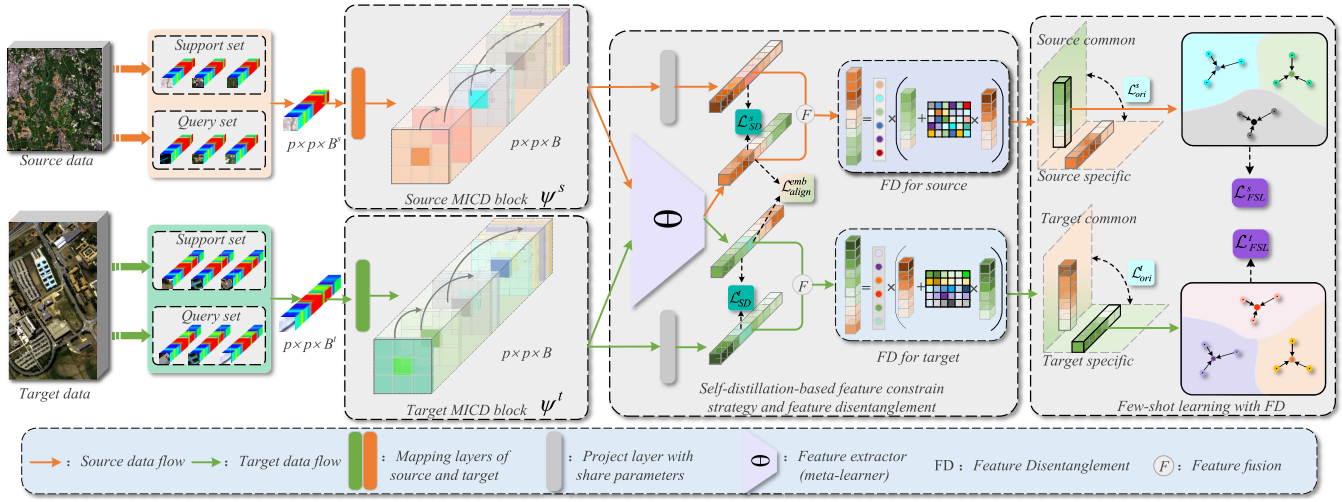

Fig. 1. Overview of the proposed FDFSL, which consists of three main components. The MICD block projects heterogeneous data with different spectral bands into the same feature space and fully explores the shared information among them in a multiorder interaction and task-driven manner. The self-distillation constraint strategy aims to enrich the feature representation for subsequent feature disentanglement tasks and prevent the model from overfitting to the domain alignment task. FSL with feature disentanglement attempts to suppress the knowledge biased toward the source domain and encourage the model to learn specific information from the target domain.

metatask constructed by domain-common features will effectively reduce the bias from the source domain, while the target metatask constructed by domain-specific features enables the model to focus on target-specific attributes implicitly.

III. Method

The overall framework of the proposed FDFSL is illustrated in Fig. 1. First, the source and target HSIs are reorganized in the episode format. Then, the source and target MICD can project the respective features into the same-dimensional feature space and emphasize important bands through multiorder interactions. Next, a parameter-shared feature extractor (i.e., metalearner) is used to obtain embedding features, and these features are constrained by interacting the shallow feature in a self-distillation manner. Finally, the features output from self-distillation are decoupled into domain-common and domain-specific components, and the required components are selectively utilized for metalearning based on the attributes of each domain.

A. Multiorder Spectral Interaction Module Based on Central Position Encoding

For the sake of spatial information, assuming that n^s source samples $\mathbf{X}^s = \{\mathbf{x}_i^s\}_i^{n^s} \in \mathbb{R}^{p\times p\times B^s}$ with C^s classes are sampled from the source hyperpsectral dataset and patched with surroundings, while n^t target patched samples $\mathbf{X}^t = \{\mathbf{x}_i^t\}_i^{n^t} \in \mathbb{R}^{p\times p\times B^t}$ with C^t classes are derived from the target hyperpsectral dataset, where $C^s > C^t$, $n^s \gg n^t$, p is the size of spatial patches, B^s and B^t are the numbers of the spectral bands of the source and target scenes, respectively. Obviously, there is naturally a heterogeneity of feature spaces between B^s-dimensional $\mathbf{X}^s$ and B^t-dimensional $\mathbf{X}^t$, resulting in the incapacity of directly measuring the distribution discrepancy between the two types of data. Even worse, due to the disjoint label spaces between source and target domains, the transferable knowledge shared between them is extremely limited.

Accordingly, mining task-relevant information among heterogeneous data, i.e., preserving share information during the common mapping of the heterogeneous feature space, is crucial for downstream FSL tasks. Previous methods commonly employ a nonshared projection layer to map heterogeneous hyperspectral data into the common feature space. However, such heterogeneous feature transformation fails to consider the share knowledge between the two domains due to nonshared network parameters. To fill this gap, we propose an MICD to emphasize informative and discriminative bands from heterogeneous data in a task-driven manner.

Specifically, as shown in Fig. 2, MICD consists of a projection layer and multiorder interaction block. Initially, the projection layer aims to map data into the B-dimensional feature space. Letting s/t represent the source or target, the embedding feature with B dimension can be obtained by $\text{Map}^{s/t}(\bullet)$. Then a multiorder interaction block is developed to fully explore the local and long-range relationships along the spectral dimension, which aims to explore the spectral correlations and emphasize task-relevant channels. To better utilize spatial structures among pixels while keeping the dominance of spectral features, we first incorporate an CPE into the features to impose spatial constraints along the spatial dimension. The initial position of pixel within hyperspectral patch $\mathbf{X}^{s/t}$ can be calculated by

$$\boldsymbol{E}_{\text{pos}}^{s/t}(x, y) = \frac{1}{|x - x_0| + |y - y_0| + 1} \tag{1}$$

where (x_0, y_0) is the position coordinates of the central pixel and (x, y) is the coordinates of the pixel within the hyperspectral patch. We first replicate $\boldsymbol{E}_{\text{pos}}^{s/t}$ along the channel axis to make it have the same dimension as the features $\mathbf{x}_i^{s/t}$, denoted as $\mathbf{E}_{\text{pos}}^{s/t}$. Then, similar to the positional encoding in the Transformer [12], the CPE is treated as the learnable position

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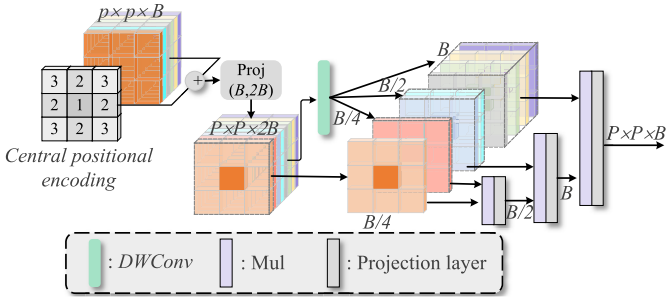

Fig. 2. Diagram of the proposed MICD block.

encoding and is added into features by

$$\boldsymbol{F}_i^{s/t} = \mathrm{Map}^{s/t}\left(\mathbf{x}_i^{s/t}\right) + \mathbf{E}_{\mathrm{pos}}^{s/t}. \tag{2}$$

An advanced theory of representation learning [19] states that deep neural networks are more favored to encode excessively low-order or high-order interactions while typically missing the most informative middle-order ones. The fashion of multiorder interaction enables the model to fully explore contextual relationships and underlying latent patterns of features. Next, the embedding feature $\boldsymbol{F}_i^{s/t}$ is first projected into a set of features with ranked dimensions (multiorder features) as follows:

$$\left[\hat{\boldsymbol{F}}_i^{s/t,b_0}, \tilde{\boldsymbol{F}}_i^{s/t,b_0}, \ldots, \tilde{\boldsymbol{F}}_i^{s/t,b_{M-1}}\right] = \rho\left(\boldsymbol{F}_i^{s/t}\right) \in \mathbb{R}^{p^2 \times b_m} \tag{3}$$

where $b^m = B/2^{(M-m-1)}$ and M is the number of interaction orders and $M > 1$, and it will degrade to self-attention when $M = 2$. It is important to note that $\hat{\boldsymbol{F}}_i^{s/t,b_0}$ and $\tilde{\boldsymbol{F}}_i^{s/t,b_0}$ are two zero-order interaction vectors with the same dimension. For ease of subsequent formula expressions, we noted them as $\hat{\boldsymbol{F}}_i^{s/t,b_0}$ and $\tilde{\boldsymbol{F}}_i^{s/t,b_0}$. Finally, the multiorder interaction is performed in a recursively gated manner

$$\hat{\boldsymbol{F}}_i^{s/t,m+1} = \phi_m\left(\tilde{\boldsymbol{F}}_i^{s/t,m}\right) \odot \rho_m\left(\hat{\boldsymbol{F}}_i^{s/t,m}\right), \quad m = 0, 1, \ldots, M-1 \tag{4}$$

where $\{\phi_m(\cdot)\}$ are a set of depthwise convolution layers and $\{\rho_m(\cdot)\}$ are projection layers used to match the dimension in different orders. After carrying out the M-order interaction along the spectral dimension, the correlation between spectral channels and the underlying pattern of global spectral features are fully explored. There are significant differences between the MICD module and channel attention mechanism (CAM). CAM primarily reinforces salient channels by employing spatial averages as attention weights. In contrast, the core of MICD lies in multiorder interactions between channels. Through recursive depth interactions, it better learns intrinsic patterns between channels, thereby enhancing task-relevant spectral bands.

B. Self-Distillation-Based Feature Constrain Strategy and Feature Disentanglement

FSL aims to realize the classification task with few-shot labeled samples per class on the target dataset, known as C-way K-shot task $\mathcal{T}$, by leveraging the metaknowledge learned from the metatask $\mathcal{T}^s$ built on source dataset with numerous labeled data. For the cross-domain FSL, the source dataset and target dataset are typically drawn from different distributions, denoted as $\mathcal{D}^s$ and $\mathcal{D}^t$, respectively. Following the common setups in prior hyperspectral FSL works, two cross-domain FSL tasks, $\mathcal{T}^s$ and $\mathcal{T}^t$, are reorganized via $\mathbf{X}^s$ and $\mathbf{X}^t$. First, in the episodic training procedure, C classes with K-labeled samples per class are randomly selected as support set $\mathcal{S}^{s/t} = \{(\mathbf{x}_i^{s/t}, \mathbf{y}_i^{s/t})\}_{i=1}^{C\times K}$, and Q samples are chosen to form the query set $\mathcal{Q}^{s/t} = \{(\mathbf{x}_i^{s/t}, \mathbf{y}_i^{s/t})\}_{i=C\times K+1}^{C\times K+Q}$, where $\mathbf{y}_j^{s/t}$ is the labels and is used for the evaluation of each episode. Due to the few labeled samples in $\mathbf{X}^t$, we employed a noisy data augment scheme to enlarge the target labeled dataset referring to the previous work [32], [33] so that a metatask is constructed.

Denoted the $\psi^{s/t}(\bullet)$ as the MICD block of the source branch or target branch. Then the metaknowledge can be learned by feeding interacted features into a parameters-share residual 3-D CNN network $\theta(\bullet)$ (i.e., feature extractor), which consists of two 3-D residual blocks. In the $\theta(\bullet)$ network, each residual block consists of three 3-D convolutional layers with a kernel size of $3 \times 3 \times 3$, skip connection layers, and two maximum pooling layers. The output embedding feature can be obtained by

$$\boldsymbol{f}^{s/t,\mathrm{emb}} = \theta\left(\psi^{s/t}\left(\mathbf{X}^{s/t}\right)\right) \in \mathbb{R}^{B^{\mathrm{emb}}}. \tag{5}$$

In order to enhance the representational capacity and stable the training of FDFSL, the interacted shallow feature $\boldsymbol{F}^{s/t,\mathcal{M}}$ is leveraged to constrain the embedding feature $\boldsymbol{f}^{s/t,\mathrm{emb}}$ and enrich the feature representation in the latent space by using a self-distillation strategy. Notably, differing from online distillation where the teacher and student models are two heterogeneous network architectures, the teacher and student models in self-distillation originate from different layers of the same network. This is because the primary aim of self-distillation is to transfer knowledge from one part of the network to another, such as conveying deep knowledge to shallow layers. In this way, it can reduce the overfitting of features in deep latent space. In addition, self-distillation is primarily focused on the transfer of knowledge, unlike contrastive learning or attention-based residual block, which directly learns the embedding space for pairs of instances. Particularly, the self-distillation-based feature constraint strategy will bring about two benefits: 1) it can prevent the model from being overfit to the domain alignment task via the guidance (i.e., Kullback–Leibler (KL) divergence) [49] between $\boldsymbol{F}^{s/t,\mathcal{M}}$ and $\boldsymbol{f}^{s/t,\mathrm{emb}}$ and 2) the fusion of $\boldsymbol{F}^{s/t,\mathcal{M}}$ and $\boldsymbol{f}^{s/t,\mathrm{emb}}$ will enrich the feature representation and facilitate the disentanglement of the desired feature components. Specifically, the interacted shallow feature map $\boldsymbol{F}^{s/t,\mathcal{M}}$ of the source or target domain is first transformed into feature vectors of the same dimension as $\boldsymbol{f}^{s/t,\mathrm{emb}}$ by

$$\boldsymbol{f}^{s/t,\mathcal{M}} = \rho^{\mathcal{M}}\left(\mathrm{Avg}\left(\boldsymbol{F}^{s/t,\mathcal{M}}\right)\right) \in \mathbb{R}^{B^{\mathrm{emb}}} \tag{6}$$

where $\mathrm{Avg}(\bullet)$ denotes the global average pooling and $\rho^{\mathcal{M}}(\bullet)$ is a parameter-shared projection layer. $\boldsymbol{f}^{s/t,\mathcal{M}}$ is squeezed into the deep embedding feature $\boldsymbol{f}^{s/t,\mathrm{emb}}$ by calculating the KL

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divergence as a supervision from the distillation

$$\mathcal{L}_{\mathrm{SD}}^{s/t} = \mathbb{KL}\left(\mathbf{P}^{s/t,\mathcal{M}} || \mathbf{P}^{s/t,\mathrm{emb}}\right) = \sum_{i=1}^{C\times K}\left[\mathbf{p}_i^{s/t,\mathrm{emb}}\log\mathbf{p}_i^{s/t,\mathrm{emb}} - \mathbf{p}_i^{s/t,\mathrm{emb}}\log\mathbf{p}_i^{s/t,\mathcal{M}}\right] \quad (7)$$

where $\mathbf{p}_i^{s/t,\mathrm{emb}}$ and $\mathbf{p}_i^{s/t,\mathcal{M}}$ are the softmax outputs of $\boldsymbol{f}^{s/t,\mathrm{emb}}$ and $\boldsymbol{f}^{s/t,\mathcal{M}}$, respectively. The self-distillation aims to approximate the embedding features by the interacted features to prevent $\boldsymbol{f}^{s/t,\mathrm{emb}}$ from being overfit to the domain alignment task. Then the final output features $\boldsymbol{f}^{s/t,o}$ for the feature disentanglement can be acquired via the fusion of $\boldsymbol{f}^{s/t,\mathcal{M}}$ and $\boldsymbol{f}^{s/t,\mathrm{emb}}$, and we use the sum operator as the fusion scheme. In particular, after enriching the feature representation of $\boldsymbol{f}^{s/t,o}$, the more diverse feature space will enable the feature disentanglement to obtain the desired feature components more conveniently. Additionally, the feature extractor θ with the fusion block mainly aims to avoid embedding features overfitting to domain alignment tasks through a self-distillation scheme from the level of distribution.

Next, the output feature $\boldsymbol{f}^{s/t,o}$ will be decoupled into the domain-common and domain-specific components. Notably, the method of feature decomposition is trivial, and we adopt a simple yet effective low-rank decomposition method [50] to decouple the features. Specifically, assuming that source and target output features $\boldsymbol{f}^{s/t,o}$ after self-distillation are uniformly generated by

$$\boldsymbol{f}^{s/t,o} = \mathbf{y}^{s/t}\left(\boldsymbol{e}^{s/t,\alpha} + \epsilon^{s/t}\boldsymbol{e}^{s/t,\beta}\right) + \mathcal{N}\left(0, \Sigma^{s/t}\right) \quad (8)$$

where $\boldsymbol{e}^{s/t,\alpha}$ and $\boldsymbol{e}^{s/t,\beta}$ are the domain-common and domain-specific features, respectively, and $\boldsymbol{e}^{s/t,\alpha} \perp \boldsymbol{e}^{s/t,\beta}$. $\epsilon^{s/t}$ is the correlation coefficient between the specific feature and the label ranging from positive to negative across various domains, and $\mathcal{N}(0, \Sigma^{s/t})$ is a standard normal random variable with mean zero and covariance matrix $\Sigma^{s/t}$. Then the parameter of the domain-specific layer can be formulated as

$$\Omega^{s/t} = \Omega^{\alpha} + \mathbf{W}^{s/t,\beta}\xi^{s/t} \quad (9)$$

where $\Omega^{\alpha} \in \mathbb{R}^{C\times B^{\mathrm{emb}}}$ are common parameters, $\mathbf{W}^{s/t,\beta} \in \mathbb{R}^{C\times B^{\mathrm{emb}}\times N_d}$ are the domain-specific parameters, N_d is the latent dimension of the domain space, and $\xi^{s/t} = [\epsilon^{s/t,s}, \epsilon^{s/t,t}]^T$ represents the coefficients. Given that the cross-domain FSL task involves only two domains, i.e., $N_d = 2$, we employ full-rank parameters. Moreover, to facilitate subsequent notation, we directly define the 3×3 identity matrix as $\mathbb{I}^{3\times 3}$. Then, orthonormality regularizers are employed to constraint the common and specific components as follows:

$$\mathcal{L}_{\mathrm{ori}}^{s/t} = \sum_{c=1}^{C}\left\|\mathbb{I}^{3\times 3} - \left(\hat{\mathbf{W}}_c^{s/t}\right)^T\hat{\mathbf{W}}_c^{s/t}\right\|_F^2 \quad (10)$$

where $\hat{\mathbf{W}}^{s/t} = [(\Omega^{\alpha})^T, (\mathbf{W}^{s/t,\beta})^T]^T$. For domain-common knowledge, it predominantly represents patterns shared between the source and target domains, derived from shared features exhibiting similar statistical properties. As for domain-specific knowledge, it mainly originates from domain-specific features, emphasizing knowledge and attributes unique to each domain.

C. Cross-Domain FSL Based on Decoupled Features and Overall Optimization

Conventional CDFSL methods typically leverage $\boldsymbol{f}_i^{s/t,\mathrm{emb}}$ to perform the episodic metric, where the massive source data may cause the model to bias toward the characteristic attribution of the source domain. In FDFSL, the FSL is performed after decoupling the source and target features. Specifically, the disentangled features $\{\mathbf{e}_i^{s,\alpha}, \mathbf{e}_i^{s,\beta}\}$ and $\{\mathbf{e}_i^{t,\alpha}, \mathbf{e}_i^{t,\beta}\}$ are selectively leveraged to perform FSL according to the task objectives, and we adopt the Euclidean distance to measure the distance between the feature of the query set and the prototype of each class, i.e.,

$$P\left(\hat{\mathbf{y}}_i^s|\mathbf{x}_i^s\right) = \mathrm{Softmax}\left(-D_{\mathrm{Euc}}\left(\mathbf{e}_i^{s,\alpha}, \mathbf{p}_c^{s,\alpha}\right)\right)$$
$$P\left(\hat{\mathbf{y}}_i^t|\mathbf{x}_i^t\right) = \mathrm{Softmax}\left(-D_{\mathrm{Euc}}\left(\mathbf{e}_i^{t,\beta}, \mathbf{p}_c^{t,\beta}\right)\right) \quad (11)$$

where $\mathbf{e}_i^{s,\alpha}$ and $\mathbf{e}_i^{t,\beta}$ are the source-common and target-specific components of sample $\mathbf{x}_i$, respectively, $D_{\mathrm{Euc}}(\bullet, \bullet)$ is the Euclidean distance, and $\mathbf{p}_c^{s/t,\alpha/\beta}$ represents the source or target prototype of class c. The FSL loss of a query sample can be calculated as follows:

$$\mathcal{L}_{\mathrm{FSL}}^{s/t}\left(P\left(\hat{\mathbf{y}}_i^{s/t}\mathbf{x}_i^{s/t}\right), \quad \mathbf{y}_i^{s/t}\right) = -\mathbf{y}_i^{s/t}\log P\left(\hat{\mathbf{y}}_i^{s/t}|\mathbf{x}_i^{s/t}\right). \quad (12)$$

Finally, the overall optimization of FDFSL is given by

$$\mathcal{L}^{s/t} = \mathcal{L}_{\mathrm{FSL}}^{s/t} + \lambda_1\left(\mathcal{L}_{\mathrm{align}}^{\mathrm{emb}} + \mathcal{L}_{\mathrm{ori}}^{s/t}\right) + \lambda_2\mathcal{L}_{\mathrm{SD}}^{s/t} \quad (13)$$

where $\mathcal{L}_{\mathrm{align}}^{\mathrm{emb}}$ is a deep coral loss to narrow the distribution gap of the source and target embedding features, and λ_1 and λ_2 are the regularization parameters that aim to optimize the energy of domain alignment and self-distillation training, respectively. Notably, the metalearner θ is alternately optimized by the loss of the source and target FSL episodes, that is, one episode is dedicated to optimizing the source loss and the next episode for target. During the metatest stage, the metalearner θ is used to map unlabeled data into the embedding space, then a nearest neighbor classifier is applied to predict labels for the query features.

IV. Experimental Results and Analysis

In this section, the experimental datasets are first introduced. Then a comparison with various state-of-the-art is reported and the results are analyzed. Next, the implementation details and evaluation metrics are given. Finally, the parameters analysis and related ablation studies are presented and discussed.

A. Experimental Datasets

Following the benchmarks in prior works, we select the Chikusei dataset as the source scene. The Chikusei data was gathered by the Headwall Hyperspec-VNIR-C sensor on July 29, 2014, which consists of 198 spectral bands in the range from 363 to 1018 nm. The spatial size of this data is 2517 × 2335, and their pseudo color image and ground truth are shown in Fig. 3. Its ground truth contains 19 classes and the detailed descriptions are listed in Table I.

With respect to the target datasets, three benchmarks, i.e., Indian Pines, Salinas, and Pavia University, are selected to

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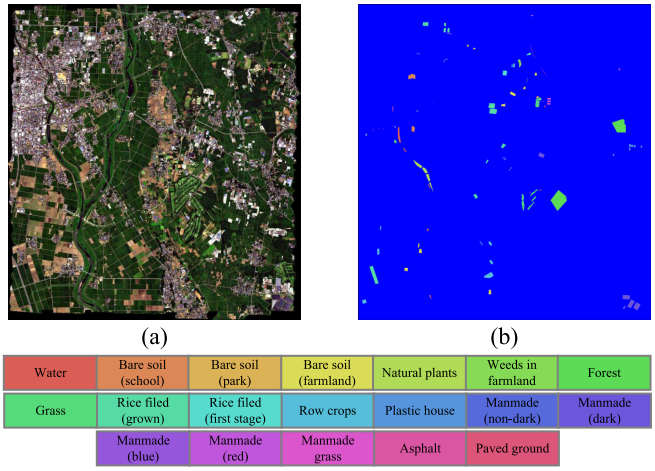

Fig. 3. Overview of the Chikusei dataset. (a) Pseudo color image of Chikusei. (b) Ground truth of Chikusei.

TABLE I
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE CHIKUSEI DATASET

No.	Land cover type	Samples
C01	Water	2345
C02	Bare soil (school)	2859
C03	Bare soil (park)	236
C04	Bare soil (farmland)	48525
C05	Natural plants	4297
C06	Weeds in farmland	1108
C07	Forest	20516
C08	Grass	6515
C09	Rice filed (grown)	13369
C10	Rice filed (first stage)	1268
C11	Row crops	5961
C12	Plastic house	2193
C13	Manmade (non-dark)	1220
C14	Manmade (dark)	7664
C15	Manmade (blue)	431
C16	Manmade (red)	222
C17	Manmade grass	1040
C18	Asphalt	801
C19	Paved ground	145
	Total	77592

TABLE II
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE INDIAN PINES DATASET

No.	Land cover type	Samples
C01	Alfalfa	46
C02	Corn-notill (school)	1428
C03	Corn-mintill	830
C04	Corn	237
C05	Grass-pasture	483
C06	Grass-trees	730
C07	Grass-pasture-mowed	28
C08	Hay-windrowed	478
C09	Oats	20
C10	Soybean-notill	972
C11	Soybean-mintill	2455
C12	Soybean-clean	593
C13	Wheat	205
C14	Woods	1265
C15	Buildings-Grass-Trees-Drives	386
C16	Stone-Steel-Towers	93
	Total	10249

TABLE III
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE SALINAS DATASET

No.	Land cover type	Samples
C01	Brocoli_green_weeds_1	2009
C02	Brocoli_green_weeds_2	3726
C03	Fallow	1976
C04	Fallow_rough_plow	1394
C05	Fallow_smooth	2678
C06	Stubble	3959
C07	Celery	3579
C08	Grapes_untrained	11271
C09	Soil_vinyard_develop	6203
C10	Corn_senesced_green_weeds	3278
C11	Lettuce_romaine_4wk	1068
C12	Lettuce_romaine_5wk	1927
C13	Lettuce_romaine_6wk	916
C14	Lettuce_romaine_7wk	1070
C15	Vinyard_untrained	7268
C16	Vinyard_vertical_trellis	1807
	Total	54129

evaluate FDFSL. The Indian Pines collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor includes 200 bands from 0.4 to 2.5 μm after removing bad spectrum. It comprises 145 $\times$ 145 pixels with 20-m/pixel spatial resolution and 16 different land covers, and the detailed sample numbers are depicted in Table II. The Salinas data is also captured by the same sensor with Indian Pines (i.e., AVIRIS). It preserves 200 bands and abandons 24 water-absorbed bands, and Salinas includes 512 $\times$ 217 pixels with a spatial resolution of 3.7 m. The sample numbers of 16 crop classes are presented in Table III. The Pavia University is acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor. The spatial size of the input data is 610 $\times$ 340, and it contains 115 spectral channels from 0.43 to 0.86 μm after removing 13 noisy spectral bands. The ground truth of Pavia University contains 42 776 labeled pixels and nine classes, which are listed in Table IV.

B. Implementation and Evaluation Metrics

The FDFSL is implemented using Python and the PyTorch framework, and all subsequent experiments are implemented on the same hardware platform with GTX-3090 GPU, Intel 4210R CPU, and 45 GB memory. For the metatraining Chikusei dataset, 200 labeled samples per class are randomly

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TABLE IV
LAND COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE UNIVERSITY OF PAVIA DATASET

No.	Land cover type	Samples
C01	Asphalt	6631
C02	Meadow	18549
C03	Gravel	2099
C04	Trees	3064
C05	Sheets	1345
C06	Bare soil	5029
C07	Bitumen	1330
C08	Bricks	3682
C09	Shadow	947
	Total	42776

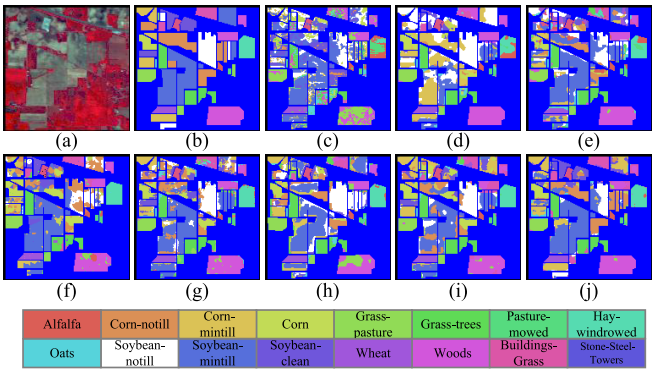

Fig. 4. Classification maps of different methods on the Indian Pines dataset. (a) Pseudo color image. (b) Ground truth. (c) SVM (40.55%). (d) SSRN (62.46%). (e) SSFTT (59.85%). (f) S3Net (61.59%). (g) DCFSL (66.70%). (h) Gia-CFSL (62.70%). (i) CMFSL (64.25%). (j) FDFSL (72.05%).

selected as $\mathbf{X}_s$ and five labeled samples per class are used as $\mathbf{X}_t$. Then, C-way K-shot metatasks are interactively constructed from the source and target datasets, and C, K are the numbers of target classes and labeled samples per target class, respectively. Following the previous settings, the target data is augmented by adding Gaussian random noise for the construction of $\mathcal{S}^t$ and $\mathcal{Q}^t$.

In terms of other parameters, the spatial size p of the hyperspectral patch is set to 9×9. The embedding dimension B_{emb} is 100 and the number of interaction orders in the MICD is set to 3. Additionally, Adam is adopted as the optimizer with learning rate of 0.001. Meanwhile, the weight decay is set to $1e^{-4}$ for the stable training of the network and the number of training episodes is set to 5000 to fully learn the prototype of each class. The overall accuracy (OA), class-specific accuracy (AA), and the Kappa coefficient are employed to evaluate the classification performance. To avoid biased estimates, all experiments were conducted with ten independent tests.

C. Comparison

To evaluate the proposed method, extensive experiments compared with other seven state-of-the-art methods are conducted on three benchmark datasets. Specifically, support vector machine (SVM) [2], spectral-spatial residual network (SSRN) [51], and spectral-spatial feature tokenization transformer (SSFTT) [52] are three supervised methods, and each based on traditional machine learning algorithms,

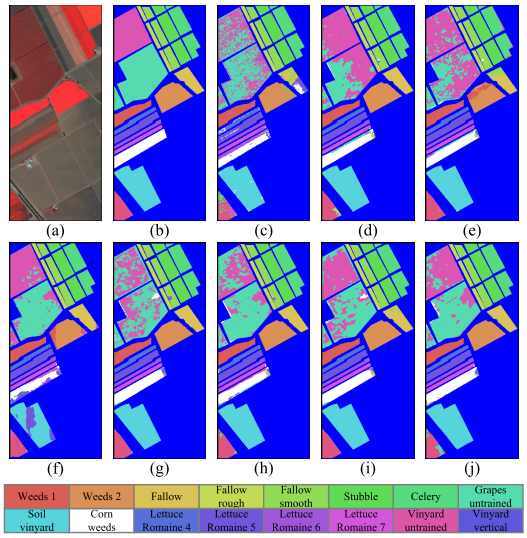

Fig. 5. Classification maps of different methods on the Salinas dataset. (a) Pseudo color image. (b) Ground truth. (c) SVM (78.72%). (d) SSRN (86.74%). (e) SSFTT (82.48%). (f) S3Net (88.57%). (g) DCFSL (89.47%). (h) Gia-CFSL (87.87%). (i) CMFSL (91.57%). (j) FDFSL (91.94%).

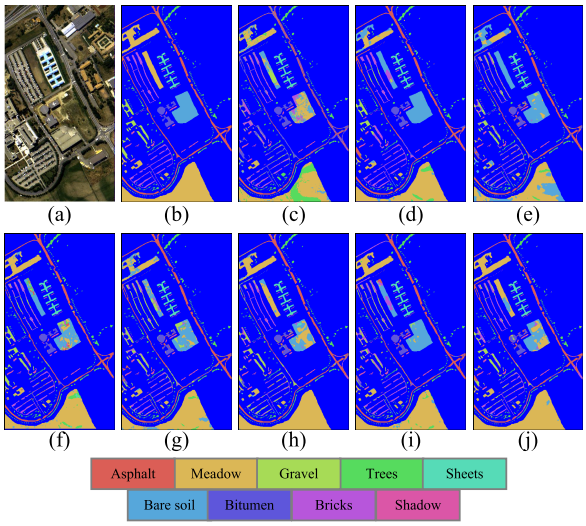

Fig. 6. Classification maps of different methods on the Pavia University dataset. (a) Pseudo color image. (b) Ground truth. (c) SVM (62.64%). (d) SSRN (84.17%). (e) SSFTT (59.85%). (f) S3Net (83.57%). (g) DCFSL (80.69%). (h) Gia-CFSL (84.53%). (i) CMFSL (85.29%). (j) FDFSL (86.61%).

CNN architecture, and Transformer architecture, respectively. Spectral-spatial siamese network (S3Net) [37] is a conventional few-shot method, where both the training and testing data are sourced only from the target domain. Deep cross-domain few-shot learning (DCFSL) [32], Gia-CFSL [33], and few-shot learning with classcovariance metric (CMFSL) [35] are all cross-domain FSL methods. Their source domain is Chikusei, and the CMFSL algorithm utilizes band selection based on optimal clustering (OCBS) [53] to select 100 bands for both the source and target domains to keep their dimensions consistent. The configuration of hyperparameters in the compared methods is assigned as the values reported in original papers. Quantitative results with five randomly labeled samples on the three datasets are reported in Tables V–VII, and classification maps are presented in Figs. 4–6. Finally, OAs for one-shot to five-shot are shown in Fig. 7.

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TABLE V
CLASSIFICATION ACCURACIES OF METHODS ON THE INDIAN PINES (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [2]	SSRN [50]	SSFTT [51]	S3Net [37]	DCFSL [32]	Gia-CFSL [33]	CMFSL [35]	FDFSL
1	61.24±9.54	77.71±7.88	63.16±11.03	79.50±12.80	79.76±10.48	77.99±8.72	75.54±7.36	**86.07±4.22**
2	69.89±13.41	48.63±17.59	71.41±15.24	64.20±13.24	84.43±8.89	**88.56±5.51**	80.14±10.67	83.09±8.28
3	35.64±8.77	62.33±27.15	60.39±14.26	55.67±22.45	61.44±8.21	62.22±7.26	63.81±15.84	**65.16±6.85**
4	89.32±8.15	94.07±6.18	**94.44±4.11**	92.80±6.00	94.09±3.35	91.61±6.25	92.56±2.82	89.29±5.67
5	99.02±0.33	**99.84±0.26**	98.84±2.22	97.71±2.62	99.63±0.38	99.18±0.81	99.66±0.65	99.40±0.61
6	17.94±13.05	**81.14±13.93**	50.59±15.29	67.35±22.63	76.79±11.18	70.22±9.77	80.55±12.39	75.11±12.61
7	81.94±13.86	94.93±6.19	86.86±5.12	**96.60±3.78**	80.49±6.68	75.20±7.83	83.07±5.57	88.66±7.90
8	57.79±17.13	73.40±18.08	59.09±14.02	59.48±17.72	68.75±10.01	71.01±13.14	74.62±10.68	**85.31±9.52**
9	**99.80±0.09**	99.77±0.22	98.82±1.88	99.95±0.07	99.34±0.71	96.85±3.97	97.94±3.13	94.62±3.33
OA(%)	63.06±4.12	67.17±7.54	69.67±5.95	71.01±4.88	81.71±3.83	82.28±1.91	80.19±4.09	**83.31±1.89**
AA(%)	68.07±3.12	81.31±3.50	75.95±3.58	79.25±2.86	82.75±2.10	81.43±1.80	83.10±1.58	**85.19±2.23**
Kappa	52.87±4.14	60.72±7.66	61.58±6.70	63.93±5.12	76.41±4.48	76.85±2.27	74.72±4.67	**78.34±2.16**

TABLE VI
CLASSIFICATION ACCURACIES OF METHODS ON THE SALINAS (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [2]	SSRN [50]	SSFTT [51]	S3Net [37]	DCFSL [32]	Gia-CFSL [33]	CMFSL [35]	FDFSL
1	76.30±12.58	96.58±4.90	97.32±3.17	97.80±3.35	95.61±7.05	85.51±13.61	**98.54±1.95**	96.59±2.72
2	22.72±13.45	36.42±10.89	29.54±13.75	35.70±11.07	44.43±10.20	47.67±10.76	44.71±10.93	**55.62±9.97**
3	28.23±20.85	50.64±16.85	31.41±10.44	40.35±10.66	46.47±11.12	52.80±11.72	56.48±8.83	**64.05±7.26**
4	25.99±11.33	79.09±17.21	59.53±13.03	72.80±13.99	71.51±21.10	75.07±13.87	80.73±18.28	**87.07±7.78**
5	25.55±23.36	73.43±7.16	56.80±17.38	71.26±20.24	72.57±6.79	69.19±16.00	**78.18±8.36**	75.17±12.50
6	40.82±14.09	**91.28±3.82**	84.95±8.96	70.54±21.57	83.78±6.61	79.25±8.53	87.57±6.70	83.52±10.22
7	96.43±0.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.57±1.30	98.10±3.16	**100.00±0.00**	99.57±1.30
8	25.69±11.30	78.29±17.20	67.15±13.96	73.59±17.85	80.47±12.79	84.09±10.30	**87.21±12.56**	83.11±14.36
9	83.50±10.01	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.33±2.00	**100.00±0.00**	**100.00±0.00**
10	36.84±12.28	63.38±9.74	56.38±10.81	68.51±10.50	59.18±10.85	61.93±16.15	60.07±8.79	**66.10±13.21**
11	51.20±19.66	50.37±18.67	46.77±12.77	47.82±13.13	58.86±14.10	58.58±9.25	55.03±15.03	**64.51±6.05**
12	15.72±12.56	46.94±9.94	31.92±8.89	54.47±14.69	41.68±13.50	44.02±13.92	54.17±17.88	**59.29±8.11**
13	96.29±2.66	98.70±1.79	97.15±6.74	99.25±0.84	95.85±6.11	95.46±6.06	**98.90±1.04**	98.70±1.35
14	68.69±23.72	91.40±5.11	**92.22±3.73**	75.70±16.06	82.18±13.13	83.62±6.86	83.51±7.74	88.60±5.47
15	15.44±9.97	36.17±17.69	38.95±19.27	73.44±15.59	67.17±9.10	67.65±17.15	76.25±11.45	**80.21±11.64**
16	87.31±3.93	99.09±1.67	98.64±2.48	98.98±1.72	99.09±1.42	98.56±3.37	**99.66±1.02**	98.18±1.98
OA(%)	40.58±5.38	61.73±4.18	54.78±2.35	58.63±2.77	63.17±2.80	64.44±3.21	65.54±3.49	**71.11±2.19**
AA(%)	49.80±2.34	74.49±2.74	68.04±2.13	73.76±2.13	74.90±2.09	75.05±3.23	78.81±1.33	**81.27±1.84**
Kappa	33.07±4.91	57.14±4.31	49.37±2.43	53.90±2.82	58.49±2.94	59.97±3.67	61.30±3.23	**67.55±2.44**

Table V presents the results of all methods on the Indian Pines dataset. It is evident that the proposed method has a significant performance margin over the other comparative methods. For example, FDFSL outperforms all counterparts and surpasses the second-best method by 5.57%, 2.46%, and 6.25% of OA, AA, and Kappa, respectively. It is worth pointing out that the proposed method outperforms others significantly in some challenging categories, such as the 2nd, 11th, and 12th categories. The abovementioned t-distributed stochastic neighbor embedding (t-SNE) visualization also indicates that the third and tenth categories of Indian Pines are almost completely overlapped, but FDFSL still manages to extract discriminative representations from them. The excellent performance of FDFSL may be attributed to the metalearning based on feature decomposition, which avoids the model's bias toward the source domain and enhances the modeling capability in dealing with the target domain. As a result, the model can better acquire metaknowledge from few-shot tasks.

Table VI reports on the results of all methods on the Salinas dataset. It can be observed that the FDFSL method still achieves the best performance in terms of OA, AA, and Kappa. Although the advantages of FDFSL over CMFSL are not statistically significant, FDFSL clearly outperforms CMFSL in the challenging eighth class (i.e., exceeds the second-best method by 2.07%). Furthermore, the variance of our method is generally lower than that of other methods. Taking the eighth class as an example, other methods have $\geq\pm 7\%$ standard deviation (CMFSL is 10.09%), but FDFSL only has 3.97%, suggesting that the FDFSL method is more robust and stable.

In terms of the Pavia University dataset, the FDFSL method also yields a significant advantage over other approaches as shown in Table VII. Specifically, the FDFSL outperforms the second-best counterpart by 1.03%, 2.09%, and 1.49% in OA, AA, and Kappa, respectively. Moreover, the standard deviation of FDFSL is lower than those of other comparative methods. As can be observed from the previous t-SNE analysis, the first and eighth classes in Pavia University are almost mixed completely, but the FDFSL method is still capable of distinguishing them effectively and noticeably outperforms other methods on quantitative evaluation metrics. This suggests that the FDFSL approach is capable of enhancing the metalearner's modeling ability under few-shot conditions for the extraction of discriminative representations.

Figs. 4–6 show the classification result maps of all methods on the three datasets. Obviously, the classification result maps of FDFSL are more closer to the land cover distribution of the ground truth. In particular, all methods achieve satisfactory classification maps on the Salinas dataset, whereas,

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TABLE VII

CLASSIFICATION ACCURACIES OF METHODS ON THE PAVIA UNIVERSITY (FIVE LABELED SAMPLES PER CLASS)

Class	SVM [2]	SSRN [50]	SSFTT [51]	S3Net [37]	DCFSL [32]	Gia-CFSL [33]	CMFSL [35]	**FDFSL**
1	99.39±0.48	90.41±21.39	90.65±14.53	90.19±29.42	98.74±1.11	**99.56±0.55**	97.49±4.48	98.96±1.63
2	65.10±26.14	**99.96±0.10**	97.61±1.50	98.91±1.13	99.74±0.26	98.51±0.97	99.77±0.26	98.78±3.38
3	46.10±10.70	91.57±11.49	89.48±13.24	90.99±9.07	92.48±7.40	89.96±9.27	**97.91±1.63**	89.10±1.25
4	96.96±0.43	**99.69±0.32**	98.59±1.57	99.73±0.33	99.43±0.56	98.99±0.87	96.37±2.21	99.32±0.55
5	96.82±3.20	**94.20±2.88**	92.74±4.95	95.20±3.83	91.99±3.58	87.10±8.77	90.02±3.82	90.47±6.25
6	98.42±0.85	**99.91±0.15**	99.27±1.80	99.41±0.69	99.46±0.66	98.64±1.23	98.96±1.15	98.41±2.57
7	99.35±0.17	99.94±0.15	98.67±1.47	98.20±4.42	98.96±1.87	96.77±8.55	99.55±0.32	**99.76±1.31**
8	58.41±23.52	49.17±21.81	60.31±24.33	70.65±10.76	73.85±12.01	76.78±7.68	76.96±10.09	**78.85±3.97**
9	97.81±0.82	98.98±1.19	98.99±1.40	89.81±6.69	**99.26±1.04**	98.24±2.32	98.37±2.11	97.98±1.80
10	22.09±25.74	**87.98±8.87**	79.65±9.27	67.09±14.45	86.04±5.45	79.75±10.57	86.09±7.34	85.40±6.40
11	87.01±4.22	95.65±4.03	93.40±6.09	89.72±9.28	97.86±1.47	97.05±2.34	**94.87±1.17**	95.51±6.65
12	98.42±0.94	98.12±3.59	98.58±3.15	87.20±14.63	99.60±0.40	99.31±1.00	99.60±0.70	**99.64±0.50**
13	98.93±0.13	**99.96±0.10**	99.41±0.66	91.17±7.71	98.63±2.32	97.77±1.19	96.75±1.94	98.29±1.23
14	87.71±1.58	96.85±3.52	93.18±3.87	92.65±7.29	98.95±0.81	97.93±1.38	96.23±2.30	**99.00±0.93**
15	50.04±23.96	81.10±18.09	57.98±23.01	78.04±9.57	75.21±7.19	74.51±7.22	79.93±7.04	**82.53±6.88**
16	72.78±10.76	89.75±7.31	85.45±20.44	85.24±7.64	88.27±7.39	87.68±6.95	92.01±8.03	**92.04±8.10**
OA(%)	73.39±3.16	84.49±3.45	82.49±2.83	85.13±3.78	88.94±2.11	88.29±1.16	90.09±2.14	**90.51±1.17**
AA(%)	79.71±2.17	92.08±1.94	89.62±2.70	89.01±3.48	93.65±0.90	92.42±0.97	93.87±1.14	**93.94±1.26**
Kappa	70.51±3.32	82.87±3.76	80.56±3.09	83.55±4.12	87.73±2.31	86.98±1.28	89.12±1.90	**89.45±1.30**

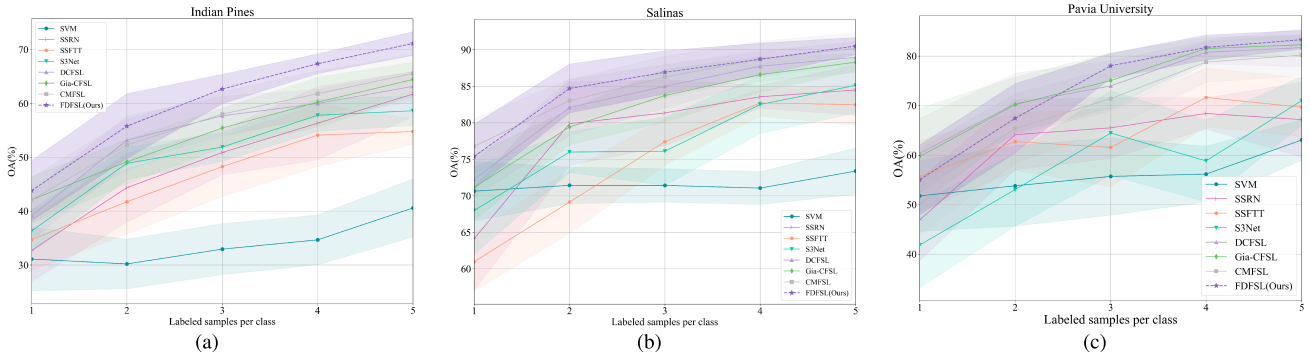

Fig. 7. OAs of different methods on the three datasets with different numbers of labeled samples. (a) Indian Pines. (b) Salinas. (c) Pavia University.

TABLE VIII

ABLATION STUDY ON THE THREE DATASETS

Dataset	Method	OA(%)	AA(%)	Kappa×100
Indian Pines	w/o FSL	52.62±3.09	65.02±2.70	47.20±3.26
	w/o FD	67.12±3.02	78.16±1.84	63.08±3.13
	w/o MICD	67.93±2.42	78.91±1.98	63.95±2.60
	w/o Self-di.	67.60±2.76	78.42±1.66	63.58±3.03
	FDFSL	**71.11±2.19**	**81.27±1.84**	**67.55±2.44**
Salinas	w/o FSL	82.04±1.87	86.78±1.14	80.08±2.04
	w/o FD	89.41±1.76	93.75±1.07	88.24±1.94
	w/o MICD	89.71±1.65	93.57±1.16	88.57±1.82
	w/o Self-di.	90.12±1.40	93.87±0.90	89.02±0.90
	FDFSL	**90.51±1.17**	**93.94±1.26**	**89.45±1.30**
Pavia University	w/o FSL	63.33±3.42	66.42±3.57	53.73±3.92
	w/o FD	82.01±2.67	83.25±2.46	76.61±3.27
	w/o MICD	79.34±1.88	79.96±2.90	73.20±2.25
	w/o Self-di.	82.99±2.57	85.00±2.37	77.93±2.99
	FDFSL	**83.31±1.89**	**85.19±2.23**	**78.34±2.16**

on the IP dataset, many compared methods do not have very good performance, especially for supervised methods. This is mainly because of the limitation of the number of labeled samples, which makes models fail to observe the global data distribution. Moreover, the Indian Pines dataset contains more intractable noise and spectral uncertainty problem, further hindering the models from achieving desired classification maps. In contrast, FDFSL generates clearer classification maps and achieves fewer misclassifications on the three datasets, yielding the superiority of FDFSL.

Finally, the OAs of different methods from one-shot to five-shot are calculated, which is shown in Fig. 7. As expected, the proposed FDFSL consistently ranks the first in almost all cases. Moreover, the FDFSL exhibits smaller standard deviations compared to other methods, and its OAs improve as the number of labeled samples becomes larger, verifying the scalability and robustness of our approach.

D. Ablation Study and Visualization

1) Ablation Comparison of Single Component: To explore the performance gains of different components in the FDFSL, it is necessary to design ablation studies for analysis of key components. It is recalled from the introduction of our framework that the key components of FDFSL mainly include FSL based on the feature disentanglement technique, MICD module, and self-distillation-based constraint strategy. Keeping other settings the same, four variants of FDFSL, i.e., the framework without FSL, feature disentanglement, MICD, and self-distillation scheme, are adopted in the ablation studies. The results of all ablation performance with five target labeled

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TABLE IX
SEQUENTIAL ABLATION STUDY ON THE THREE DATASETS

Dataset	FSL	MICD	FD	Self-di.	OA(%)	AA(%)	Kappa×100
Indian Pines	×	×	×	×	52.62±3.09	65.02±2.70	47.20±3.26
	✓	×	×	×	65.43±2.52	76.34±1.53	61.15±2.61
	✓	✓	×	×	67.07±2.90	77.95±1.60	62.99±2.99
	✓	✓	✓	×	70.82±2.01	81.21±1.56	67.33±2.14
	✓	✓	✓	✓	**71.11±2.19**	**81.27±1.84**	**67.55±2.44**
Salinas	×	×	×	×	82.04±1.87	86.78±1.14	80.08±2.04
	✓	×	×	×	88.22±1.19	92.82±0.75	86.90±1.30
	✓	✓	×	×	89.46±1.67	93.42±0.90	88.28±1.84
	✓	✓	✓	×	90.28±1.29	**94.12±0.84**	89.20±1.41
	✓	✓	✓	✓	**90.51±1.17**	93.94±1.26	**89.45±1.30**
Pavia University	×	×	×	×	63.33±3.42	66.42±3.57	53.73±3.92
	✓	×	×	×	80.71±3.10	80.36±2.35	74.88±3.81
	✓	✓	×	×	82.14±2.05	83.60±2.06	76.77±3.35
	✓	✓	✓	×	82.90±2.76	84.39±2.06	77.75±2.48
	✓	✓	✓	✓	**83.31±1.89**	**85.19±2.23**	**78.34±2.16**

samples are summarized in Table VIII, where the best results are highlighted in bold.

It can be observed from Table VIII that the FDFSL achieves the best classification performance, and removing different portions results in varying degrees of performance degradation. Specifically, the accuracies of the variants without FSL are lowest on the three datasets, indicating that FSL exhibits significant performance advantages compared to conventional supervised learning when dealing with limited labeled samples. After removing the MICD module that explores discriminative information in heterogeneous data, the OAs of FDFSL drop by 3.18%, 0.8%, and 3.97% on the three datasets, respectively. This highlights the significance of exploring discriminative bands from heterogeneous data for cross-sensor data. Particularly in the Pavia dataset, the performance improvements with the MICD are most notable, due possibly to the relatively abundant heterogeneous discriminative information between the source and target datasets.

Next, we explore the ablation performance of the self-distillation-based feature constraint scheme and observe that the performance gains of self-distillation are slight yet effective. Particularly, on the Indian Pines dataset, the feature constraints imposed by self-distillation and the enrichment of feature representation can effectively enhance the classification performance.

Finally, in terms of the core feature disentanglement technique, we replaced decomposed domain-common/specific features with holistic features for FSL. It can be seen that the FSL model exhibits the most severe performance degradation after removing the feature decoupling technique (i.e., 3.99%↓, 1.1%↓, and 1.3%↓ on the three datasets, respectively), suggesting the effectiveness of feature disentanglement. Overall, it can be concluded that each key component in our proposed framework is fruitful and reasonable for enhancing component performance.

2) Sequential Ablation Comparison: To further verify the effectiveness of different modules, we conducted a sequential ablation experiment. Specifically, we progressively added four modules (FSL, MICD block, feature disentanglement, and self-distillation scheme) to the baseline to explore their performance gains on the three datasets. Table IX reports the sequential ablation performance on the three datasets under the five-shot setting, with the best results highlighted in bold.

Evidently, with the addition of each new component, variants of our approach exhibit varying degrees of performance improvement. As anticipated, after incorporating all modules, FDFSL achieves the best classification performance on the three datasets. It is noteworthy that although the AA of FDFSL on the Salinas dataset is slightly lower than the variant without self-distillation, its OA and Kappa coefficient are significantly higher than other variants. This is attributed to the fact that FDFSL exhibits slightly lower classification accuracy on some categories with a smaller proportion of samples, while having superior accuracy in categories with a larger sample proportion. Overall, across the three datasets, with the addition of each component, there is a stable improvement in the method's classification performance, yielding the effectiveness of the proposed modules.

3) Effect of Different Feature Components: To further validate the effectiveness of the feature disentanglement approach in FSL, we extensively design multiple variants of FDFSL based on different feature components to fully explore the impact of feature decomposition on FSL. Specifically, we designed four different feature composition strategies: 1) *holistic* feature components for source and target domains, which can be viewed as a *baseline*; 2) domain-common feature components for both source and target domains (*Com.*+*Com.*); 3) domain-specific feature components for both source and target domains (*Spe.*+*Spe.*); and 4) domain-specific feature components for source domain and domain-common feature components for target domain (*Spe.*+*Com.*), which is the theoretically worst-case scenario.

The experimental results on the three datasets are shown in Fig. 8, which mainly uses the mean and standard deviation of OA with ten independent experiments as the evaluation metric. As expected, our argument of domain-common components for the source and domain-specific components for the target (i.e., *Com.*+*Spe.*) achieves the best performance and significantly outperforms the baseline based on *holistic* features. Particularly, when we leverage the combination of *Com.*+*Com.*, the performance of FDFSL is better than the *baseline* on the SA and PU datasets, indicating that the

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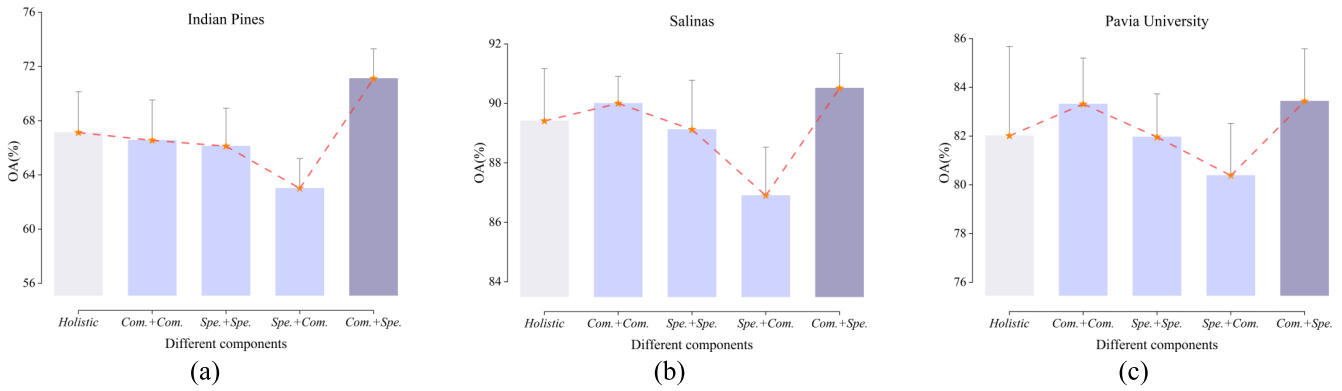

Fig. 8. Performance of FDFSL with different feature components on the three datasets. (a) Indian Pines. (b) Salinas. (c) Pavia University.

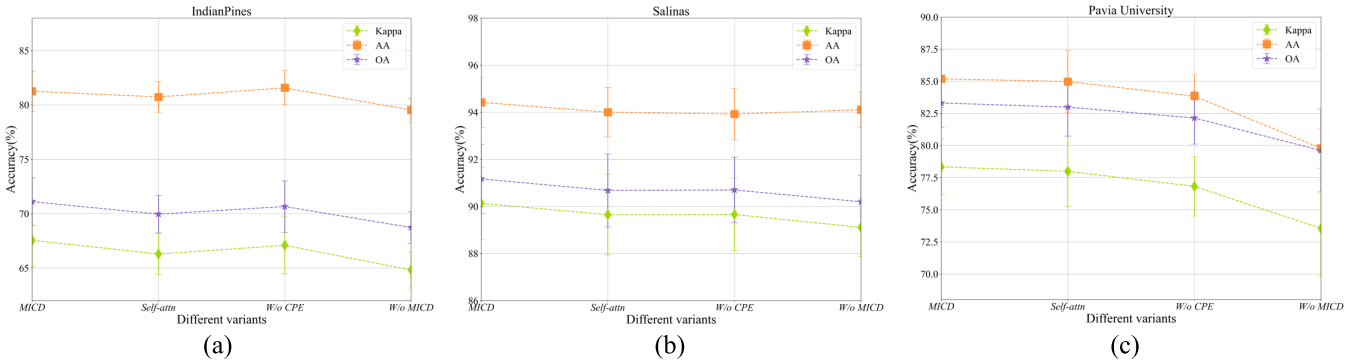

Fig. 9. Performance of FDFSL with different mapping layers on the three datasets. (a) Indian Pines. (b) Salinas. (c) Pavia University.

utilization of domain-common features will reduce domain-biased information to a certain extent and thus improve the model performance. When we adopt the theoretically worst *Spe.+Com.* combination, the model performs as expected, yielding the poorest performance, which is consistent with the theory insight. This is mainly because the source domain-specific features introduce an undeniable source domain bias while the utilization of the target domain-common features partially reduces the target-specific attributes. These two factors critically limit the adaptability of FDFSL to the target domain. In turn, when adopting our argued *Com.+Spe.* combination for FSL, FDFSL can not only reduce the model's bias toward the source domain but also implicitly guide the model to focus on the target domain.

4) Exploration of the Effectiveness of the MICD Block: In order to further explore the effectiveness of the MICD block, we designed three different variants for comparison while keeping the other components of FDFSL unchanged. Specifically, the first variant replaces the MICD block with a self-attention module (i.e., Self-attn), the second variant removes the CPE in the MICD (i.e., W/o CPE), and the MICD block is removed in the third variant (i.e., W/o MICD). From Fig. 9, it can be observed that MICD outperforms all other variants in terms of OA, AA, and Kappa. Compared to self-attention with only two-order interactions, MICD, with its multiorder recursive interaction mechanism, excels in refining task-relevant spectral bands. After removing the CPE, FDFSL yields a varying degree of performance drops on all the three datasets, suggesting the effectiveness of CPE. Finally, with the complete removal of the MICD, FDFSL's performance is the poorest. This is attributed to the simple projection lacking interchannel interactions, making it challenging to deeply explore shared and discriminative information between heterogeneous data.

5) t-SNE Visualization of FDFSL: To intuitively understand the effectiveness of the FDFSL, the original samples and embedding features $\boldsymbol{f}^{t,\text{emb}}$ of the three datasets are visualized in Fig. 10. In the case of raw hyperspectral data, due to the inherent spectral similarities among classes, some categories exhibit poor separability. For instance, the 3rd, 10th, and 13th classes of the IP dataset, the third and tenth classes of the SA dataset, and the second and sixth classes of the PU dataset nearly overlap with each other completely, which renders common linear classifiers incapable of achieving effective discrimination among them. Furthermore, some classes experience a scattered distribution due to potential spectral shifts, such as the ninth class in the IP dataset and the seventh class in the SA dataset. These obvious interclass similarity and intraclass variability pose great challenges for HSIC models. As expected, after performing the metatraining of the FDFSL framework, the metalearner is able to extract meaningful feature representations for the test data. Specifically, in the embedding feature space, the different categories are more linearly separable and tightly clustered, indicating that our learning approach can remarkably boost feature extraction capabilities of models with few labeled samples.

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TABLE X
PARAMETERS, FLOPS, AND COMPUTATIONAL TIME (SECONDS) ON EACH DATASET WITH DIFFERENT METHODS

	Methods	S3Net [37]	DCFSL [32]	Gia-CFSL [33]	CMFSL [35]	**FDFSL**
Indian Pines	#params	0.441696M	0.057696M	0.438995M	**0.613991M**	0.190550M
	FLOPs	64.652864M	43.164880M	**8868.393168M**	27.132028M	52.722785M
	Training time	9.40s	1984.93s	**4609.10s**	1196.86s	1167.07s
	Testing time	**4.16s**	1.32s	1.20s	0.69s	1.62s
Salinas	#params	0.449888M	0.058096M	0.439395M	**0.613991M**	0.190950M
	FLOPs	65.918656M	43.197280M	**8868.425568M**	27.132028M	52.755185M
	Training time	9.54s	2108.49s	**4731.29s**	1232.96s	1225.49s
	Testing time	**22.78s**	6.87s	6.18s	3.58s	9.96s
Pavia University	#params	0.238955M	0.047996M	0.428189M	**0.613991M**	0.180850M
	FLOPs	33.528484M	42.379180M	**8816.224668M**	27.132028M	51.937085M
	Training time	5.18s	1244.71s	**3023.16s**	759.21s	769.34s
	Testing time	**14.10s**	4.33s	4.90s	2.91s	6.35s

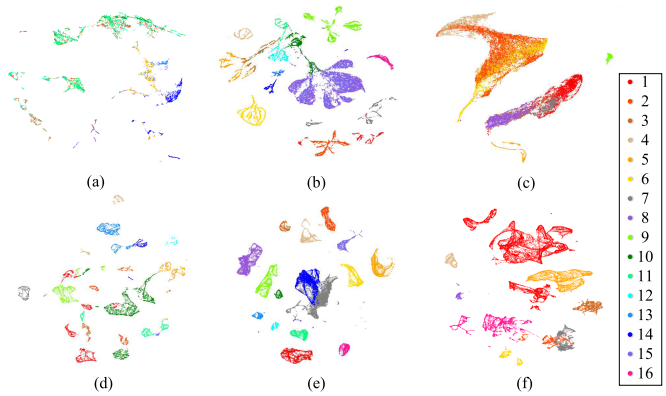

Fig. 10. t-SNE visualization of the original data and embedding features on the three datasets. The first column is the distribution of the original data. (a) Original data of Indian Pines. (b) Original data of Salinas. (c) Original data of Pavia University. The second column is the distribution of embedding features. (d) Embedding features of Indian Pines. (e) Embedding features of Salinas. (f) Embedding features of Pavia University.

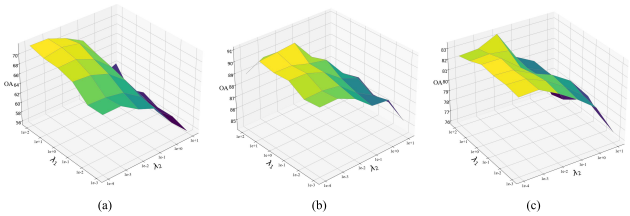

Fig. 11. Parameter analysis of λ_1 and λ_2 on the three datasets. (a) Indian Pines. (b) Salinas. (c) Pavia University.

E. Parameter Analysis and Computational Complexity

1) Parameter Analysis: The adjustable parameters of the FDFSL mainly include λ_1 and λ_2, which are responsible for optimizing the energy of the domain alignment and self-distillation training, respectively. To further investigate the sensitivity of the FDFSL to these two parameters, we carry out a parameter analysis experiment. Based on early empirical observations, it is found that the OA of the FDFSL improves with increasing λ_1 and degrades with increasing λ_2, thus the value ranges of regularization parameters are set to be $\{1e-3, 1e-2, 1e-1, 1e-0, 1e+1, 1e+2\}$ and $\{1e-4, 1e-3, 1e-2, 1e-1, 1e-0, 1e+1\}$, respectively.

The OAs of different parameter combinations on the three datasets are presented in Fig. 11, jointly indicating the model's sensitivity to λ_1 and λ_2. It can be seen in Fig. 11 that the FDFSL exhibits a significant performance degradation as λ_2 increases, while its sensitivity to λ_1 is relatively low. Specifically, when λ_1 increases from $1e^{-3}$ to $1e^{+2}$, the performance of FDFSL slightly improves and then levels off. Comprehensively, FDFSL tends to achieve a better performance when $\lambda_1 = \{1e-1, 1e-0, 1e+1\}$ on the three datasets. Regarding parameter λ_2, the performance of FDFSL noticeably decrease as λ_2 increases from $1e^{-4}$ to $1e^{+1}$, but it can achieve suboptimal performance when $\lambda_2 = \{1e-2, 1e-3, 1e-4\}$ on the three datasets. It is noteworthy that through the aforementioned ablation study, the FDFSL exhibits varying degrees of performance drops on the three datasets when the self-distillation component is removed. Overall, FDFSL is less sensitive to λ_1, while the value of λ_2 should be set relatively small. Therefore, for Indian Pines, Pavia University, and Salinas, the suboptimal parameter sets of λ_1 and λ_2 are $\{1e-1, 1e-0, 1e+1\}$ and $\{1e-2, 1e-3, 1e-4\}$, respectively. Although the two parameters exhibit consistent trends across the three datasets, there are still subtle differences in classification performance under specific configurations. This is mainly because different parameter settings will influence the optimization energy of the corresponding losses, and values that are too large or too small may lead to overfitting or underfitting of subtasks. Moreover, different datasets have distinct data distributions and biases, and this diversity in distribution may result in models with the same parameters exhibiting different implicit biases for each dataset. Hence, these factors may contribute to the subtle differences in performance observed across the three datasets for different parameter configurations.

2) Analysis of Computational Complexity: To comprehensively explore the effectiveness and limitations of our approach, we compared the computational complexity of different methods, including parameters, floating-point operations per second (FLOPS), training time, and testing time. It is important to note that we specifically compared FSL-based methods. Moreover, since methods based on cross-domain FSL involve both source and target domain data flows, for the sake of fairness in comparison, we primarily reported the computational complexity of the target data flows. This is because these methods share the same source domain dataset (i.e., Chikusei). As evident from Table X, although our method does not have the lowest computational overhead, it exhibits acceptable computational complexity, training time, and inference time.

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V. Conclusion

In this article, a novel cross-domain FDFSL was proposed for HSIC. By decoupling different feature components for FSL, FDFSL is able to suppress the metaknowledge biased toward the source domain and enhance the learning of specific information from the target domain. To the best of our knowledge, this is the first work to apply the idea of feature disentanglement to hyperspectral cross-domain FSL. In addition, to fit the inherent heterogeneous gap between source and target data caused by sensor differences, an MICD is proposed to emphasize informative and discriminative spectral bands in a task-driven manner. Finally, to prevent the model from being overfit to domain alignment tasks, a self-distillation-based feature constraint strategy is developed to incorporate semantic information into embedding features to enhance their semantic relevance and enrich the feature representation for subsequent feature disentanglement tasks. Extensive experiments and analysis were carried out to demonstrate that the proposed FDFSL outperforms seven state-of-the-art HSIC methods on three benchmark datasets, and the proposed key components showed varying degrees of improvement in model performance.

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